5 V input -> 3.3 V logic / ten isolated host supplies
ESP32-S3FN8 + W5500 -> three 74HC595 -> twenty isolated dry contacts
Twenty differential LED detectors -> twenty isolated returns -> MCU GPIO
Original case wiring bypasses this controller on passive adapters

5 V input, protection and 3.3 V buck

檔案： power.kicad_sch

ESP32-S3FN8 / USB programming (1/2)

檔案： processor_1.kicad_sch

ESP32-S3FN8 / USB programming (2/2)

檔案： processor_2.kicad_sch

W5500 / HR911105A Ethernet (1/3)

檔案： ethernet_1.kicad_sch

W5500 / HR911105A Ethernet (2/3)

檔案： ethernet_2.kicad_sch

W5500 / HR911105A Ethernet (3/3)

檔案： ethernet_3.kicad_sch

Output latches / timed drive rail (1/2)

檔案： outputs_1.kicad_sch

Output latches / timed drive rail (2/2)

檔案： outputs_2.kicad_sch

Host 01: isolated controls / LEDs (1/3)

檔案： host01_1.kicad_sch

Host 01: isolated controls / LEDs (2/3)

檔案： host01_2.kicad_sch

Host 01: isolated controls / LEDs (3/3)

檔案： host01_3.kicad_sch

Host 02: isolated controls / LEDs (1/3)

檔案： host02_1.kicad_sch

Host 02: isolated controls / LEDs (2/3)

檔案： host02_2.kicad_sch

Host 02: isolated controls / LEDs (3/3)

檔案： host02_3.kicad_sch

Host 03: isolated controls / LEDs (1/3)

檔案： host03_1.kicad_sch

Host 03: isolated controls / LEDs (2/3)

檔案： host03_2.kicad_sch

Host 03: isolated controls / LEDs (3/3)

檔案： host03_3.kicad_sch

Host 04: isolated controls / LEDs (1/3)

檔案： host04_1.kicad_sch

Host 04: isolated controls / LEDs (2/3)

檔案： host04_2.kicad_sch

Host 04: isolated controls / LEDs (3/3)

檔案： host04_3.kicad_sch

Host 05: isolated controls / LEDs (1/3)

檔案： host05_1.kicad_sch

Host 05: isolated controls / LEDs (2/3)

檔案： host05_2.kicad_sch

Host 05: isolated controls / LEDs (3/3)

檔案： host05_3.kicad_sch

Host 06: isolated controls / LEDs (1/3)

檔案： host06_1.kicad_sch

Host 06: isolated controls / LEDs (2/3)

檔案： host06_2.kicad_sch

Host 06: isolated controls / LEDs (3/3)

檔案： host06_3.kicad_sch

Host 07: isolated controls / LEDs (1/3)

檔案： host07_1.kicad_sch

Host 07: isolated controls / LEDs (2/3)

檔案： host07_2.kicad_sch

Host 07: isolated controls / LEDs (3/3)

檔案： host07_3.kicad_sch

Host 08: isolated controls / LEDs (1/3)

檔案： host08_1.kicad_sch

Host 08: isolated controls / LEDs (2/3)

檔案： host08_2.kicad_sch

Host 08: isolated controls / LEDs (3/3)

檔案： host08_3.kicad_sch

Host 09: isolated controls / LEDs (1/3)

檔案： host09_1.kicad_sch

Host 09: isolated controls / LEDs (2/3)

檔案： host09_2.kicad_sch

Host 09: isolated controls / LEDs (3/3)

檔案： host09_3.kicad_sch

Host 10: isolated controls / LEDs (1/3)

檔案： host10_1.kicad_sch

Host 10: isolated controls / LEDs (2/3)

檔案： host10_2.kicad_sch

Host 10: isolated controls / LEDs (3/3)

檔案： host10_3.kicad_sch

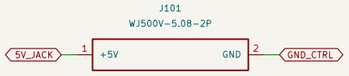
SCHEMATIC REVIEW — no PCB routing or Gerber release

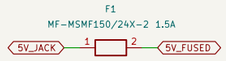
Sheet: /
File: rack-atx-controller.kicad_sch

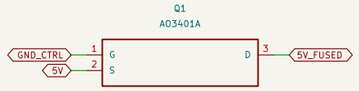
Title: 10-host rack ATX controller

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 1/39

Regulated 5 V +/−5%, 2 A, terminal 1 positive. USB VBUS is NOT a power input.
Check PCB thermal/current loops and external PSU before energizing.







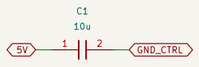


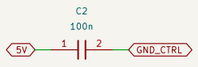




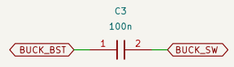


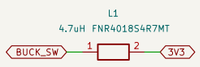


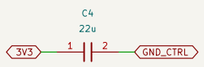














SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

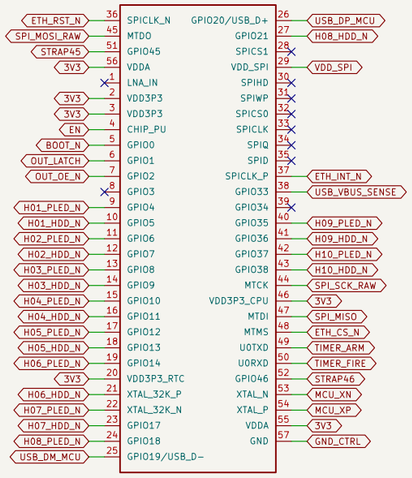
Sheet: /5 V input, protection and 3.3 V buck/
File: power.kicad_sch

Title: 5 V input, protection and 3.3 V buck

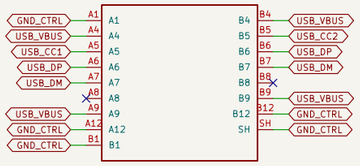
Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 2/39

Wired Ethernet only. Leave LNA_IN open and never initialize Wi-Fi/BLE.
GPIO26..32 used by internal quad flash: do not route. GPIO45 pulled low selects 3.3 V flash.

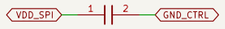
U2
ESP32-S3FN8



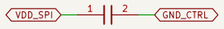
J104
TYPE-C-31-M-12



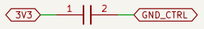
C6
100n



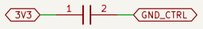
C7
1u



C8
100n



C9
100n



C10
100n



C11
100n



C12
100n

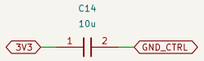


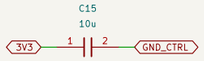
C13
100n

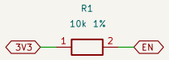


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /ESP32-S3FN8 / USB programming (1/2)/
File: processor_1.kicad_sch
Title: ESP32-S3FN8 / USB programming (1/2)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 3/39

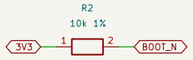
Wired Ethernet only. Leave LNA_IN open and never initialize Wi-Fi/BLE.
GPIO26..32 used by internal quad flash: do not route. GPIO45 pulled low selects 3.3 V flash.



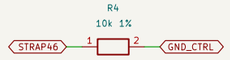


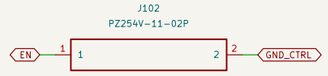




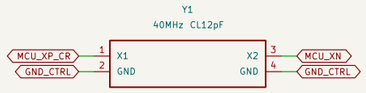


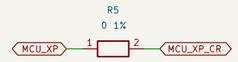




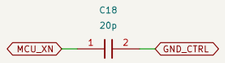


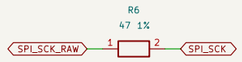










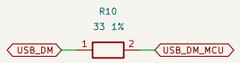


















SCHEMATIC REVIEW — not a fabrication release

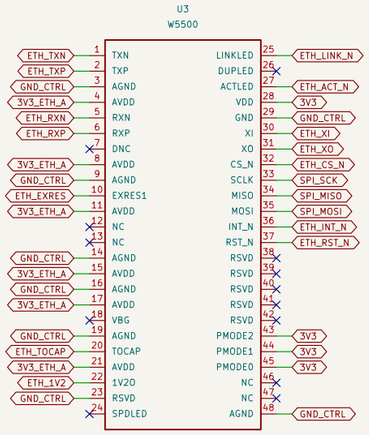
Rack ATX Controller / ddasdkimo

Sheet: /ESP32-S3FN8 / USB programming (2/2)/
File: processor_2.kicad_sch

Title: ESP32-S3FN8 / USB programming (2/2)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 4/39

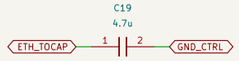
WIZnet RJ45 reference topology. HR911105A pin numbering differs from J1B1211CCD.
100-ohm differential MDI routing; crystal load caps need final frequency measurement.

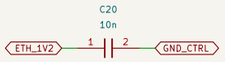


















SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /W5500 / HR911105A Ethernet (1/3)/
File: ethernet_1.kicad_sch

Title: W5500 / HR911105A Ethernet (1/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 5/39

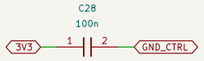
WIZnet RJ45 reference topology. HR911105A pin numbering differs from J1B1211CCD.
100-ohm differential MDI routing; crystal load caps need final frequency measurement.















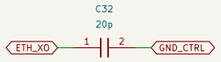
















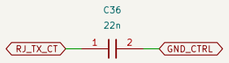




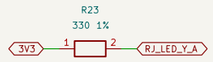


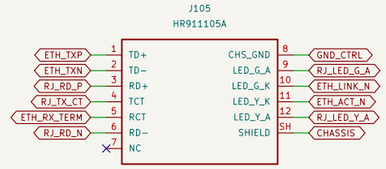












SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /W5500 / HR911105A Ethernet (2/3)/
File: ethernet_2.kicad_sch

Title: W5500 / HR911105A Ethernet (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 6/39

WIZnet RJ45 reference topology. HR911105A pin numbering differs from J1B1211CCD.
100-ohm differential MDI routing; crystal load caps need final frequency measurement.



SCHEMATIC REVIEW — not a fabrication release

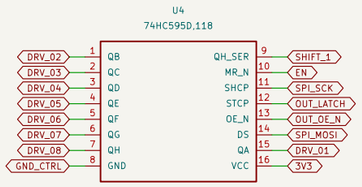
Rack ATX Controller / ddasdkimo

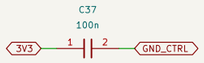
Sheet: /W5500 / HR911105A Ethernet (3/3)/
File: ethernet_3.kicad_sch

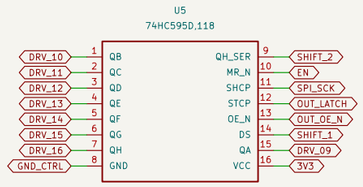
Title: W5500 / HR911105A Ethernet (3/3)

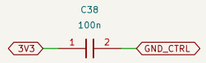
Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 7/39

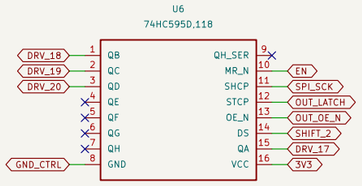
595 stores 20 output bits; unused 4 outputs not connected. Shift 24 zeros before OE low.
SPI traffic shifts internal register only; latch ONLY after a complete output frame. Never latch Ethernet data.
Timer: nominal 16.8 s, qualify 8..25 s with capacitor bias/tolerance. Firmware long press 5 s.
One command at a time. Repeated trigger edges can extend a 555 pulse: firmware is not a safety-rated system.

















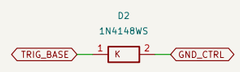




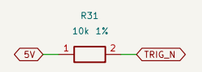




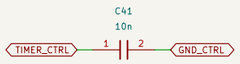


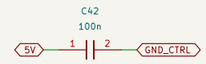






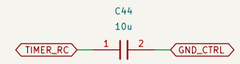
















SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /Output latches / timed drive rail (1/2)/
File: outputs_1.kicad_sch

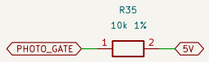
Title: Output latches / timed drive rail (1/2)

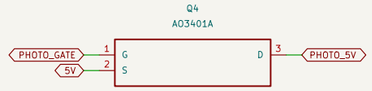
Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 8/39

595 stores 20 output bits; unused 4 outputs not connected. Shift 24 zeros before OE low.
SPI traffic shifts internal register only; latch ONLY after a complete output frame. Never latch Ethernet data.
Timer: nominal 16.8 s, qualify 8..25 s with capacitor bias/tolerance. Firmware long press 5 s.
One command at a time. Repeated trigger edges can extend a 555 pulse: firmware is not a safety-rated system.











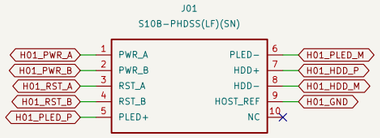


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Output latches / timed drive rail (2/2)/
File: outputs_2.kicad_sch

Title: Output latches / timed drive rail (2/2)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 9/39

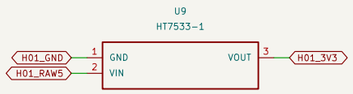
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.







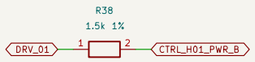






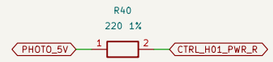




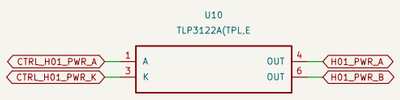














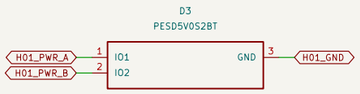


















SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 01: isolated controls / LEDs (1/3)/
File: host01_1.kicad_sch

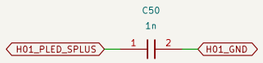
Title: Host 01: isolated controls / LEDs (1/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 10/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

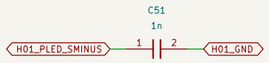
















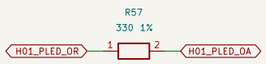


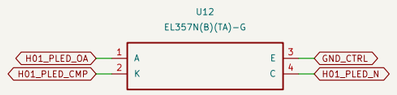










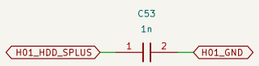




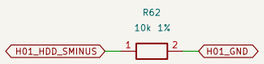


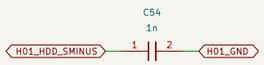














SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /Host 01: isolated controls / LEDs (2/3)/
File: host01_2.kicad_sch

Title: Host 01: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 11/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





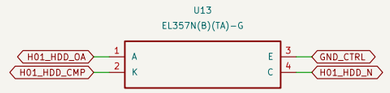






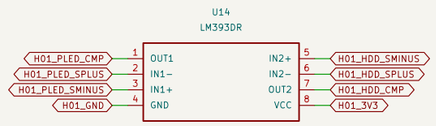










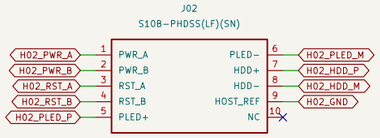


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 01: isolated controls / LEDs (3/3)/
File: host01_3.kicad_sch

Title: Host 01: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 12/39

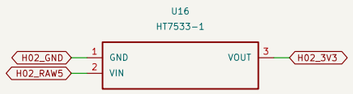
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.







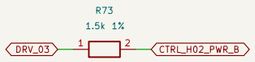










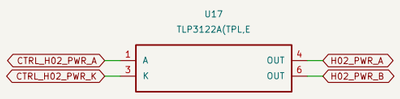




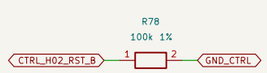












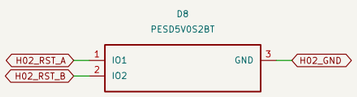
















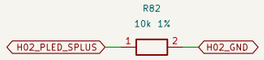
SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 02: isolated controls / LEDs (1/3)/
File: host02_1.kicad_sch

Title: Host 02: isolated controls / LEDs (1/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 13/39

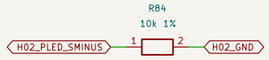
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

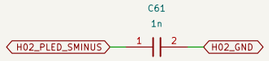




























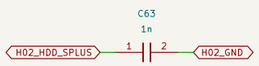






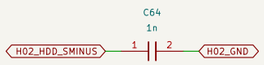














SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 02: isolated controls / LEDs (2/3)/
File: host02_2.kicad_sch

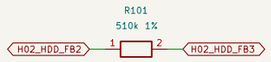
Title: Host 02: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 14/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





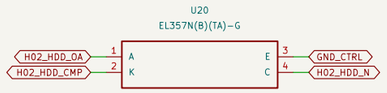






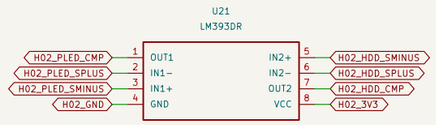










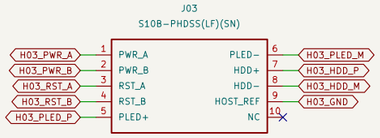


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 02: isolated controls / LEDs (3/3)/
File: host02_3.kicad_sch

Title: Host 02: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 15/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.







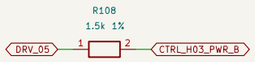


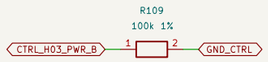






















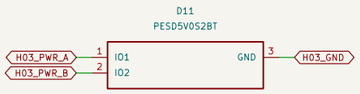


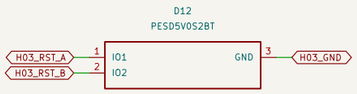




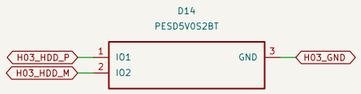












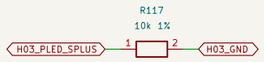
SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 03: isolated controls / LEDs (1/3)/
File: host03_1.kicad_sch

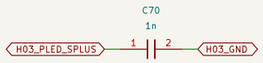
Title: Host 03: isolated controls / LEDs (1/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 16/39

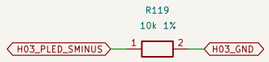
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

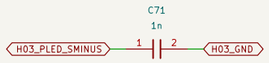


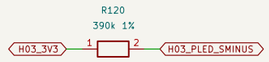
















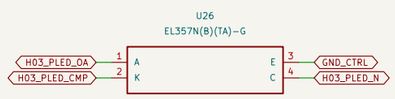










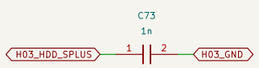




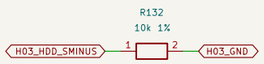


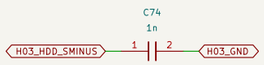


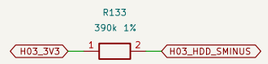












SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 03: isolated controls / LEDs (2/3)/
File: host03_2.kicad_sch

Title: Host 03: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 17/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





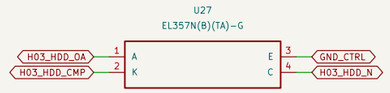






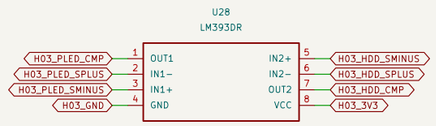










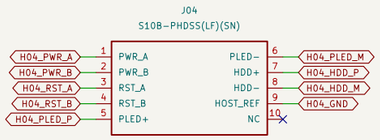


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 03: isolated controls / LEDs (3/3)/
File: host03_3.kicad_sch

Title: Host 03: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 18/39

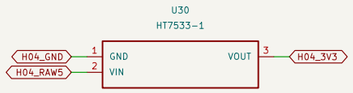
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





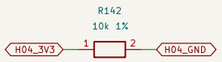


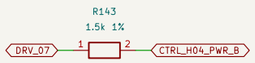


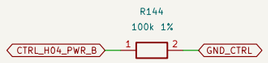






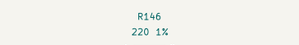


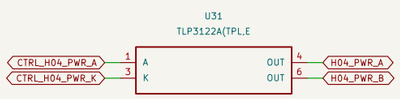














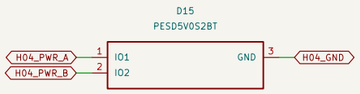


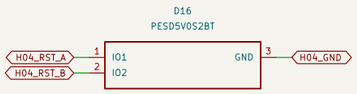




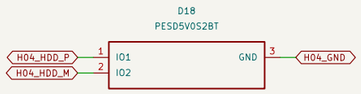








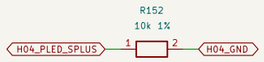


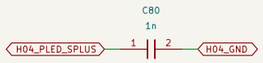


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 04: isolated controls / LEDs (1/3)/
File: host04_1.kicad_sch
Title: Host 04: isolated controls / LEDs (1/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 19/39

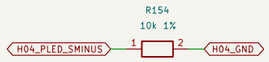
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

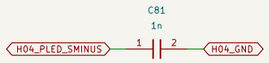
















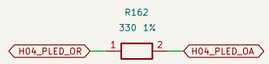




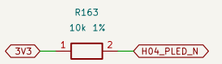








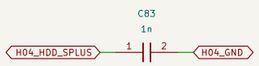




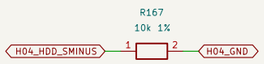


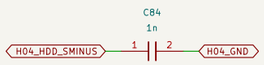














SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 04: isolated controls / LEDs (2/3)/
File: host04_2.kicad_sch
Title: Host 04: isolated controls / LEDs (2/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 20/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





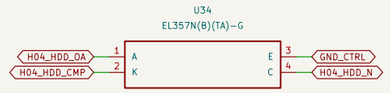




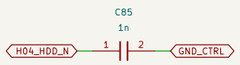


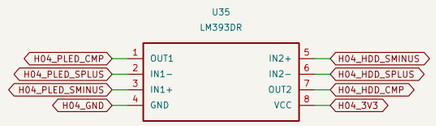










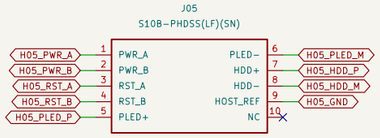


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 04: isolated controls / LEDs (3/3)/
File: host04_3.kicad_sch

Title: Host 04: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 21/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.







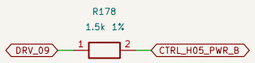
























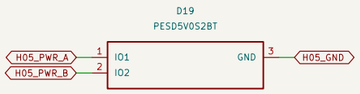














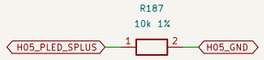




SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 05: isolated controls / LEDs (1/3)/
File: host05_1.kicad_sch
Title: Host 05: isolated controls / LEDs (1/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 22/39

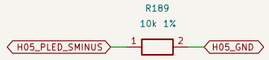
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

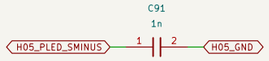


















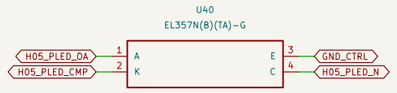










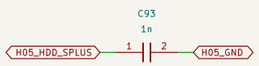




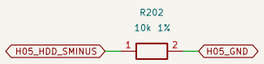


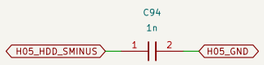


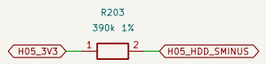












SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 05: isolated controls / LEDs (2/3)/
File: host05_2.kicad_sch
Title: Host 05: isolated controls / LEDs (2/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 23/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





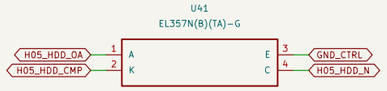


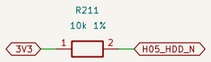




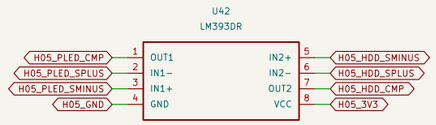










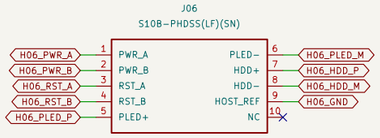


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 05: isolated controls / LEDs (3/3)/
File: host05_3.kicad_sch

Title: Host 05: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 24/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





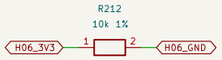


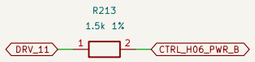


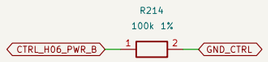






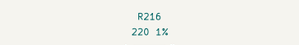


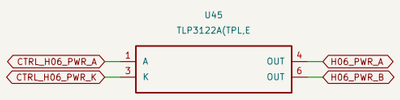








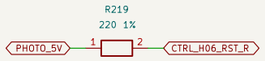




















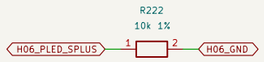


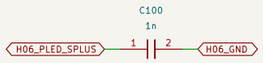


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 06: isolated controls / LEDs (1/3)/
File: host06_1.kicad_sch
Title: Host 06: isolated controls / LEDs (1/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 25/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

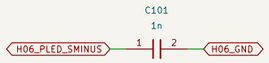
















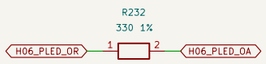


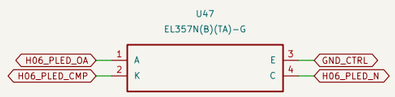


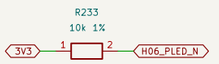


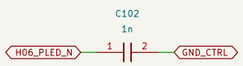






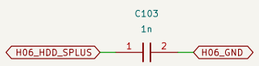




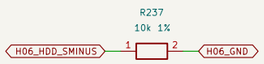


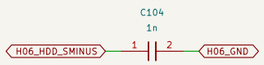


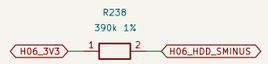












SCHEMATIC REVIEW — not a fabrication release

Rack ATX Controller / ddasdkimo

Sheet: /Host 06: isolated controls / LEDs (2/3)/
File: host06_2.kicad_sch

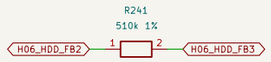
Title: Host 06: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 26/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





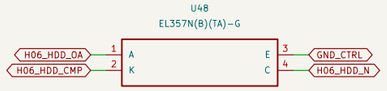




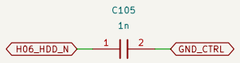












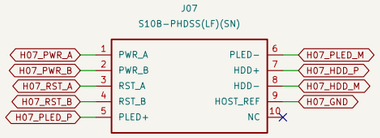


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 06: isolated controls / LEDs (3/3)/
File: host06_3.kicad_sch

Title: Host 06: isolated controls / LEDs (3/3)

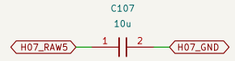
Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 27/39

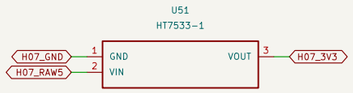
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.



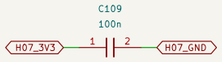




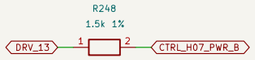














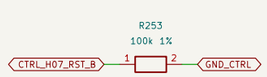
























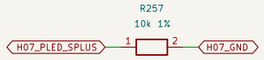


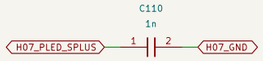


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 07: isolated controls / LEDs (1/3)/
File: host07_1.kicad_sch
Title: Host 07: isolated controls / LEDs (1/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 28/39

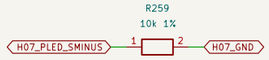
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

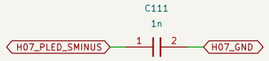


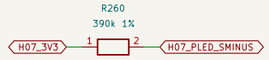


















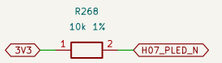








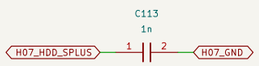




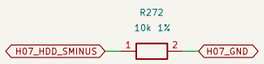


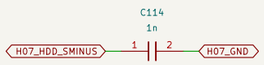


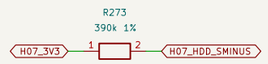












SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 07: isolated controls / LEDs (2/3)/
File: host07_2.kicad_sch

Title: Host 07: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 29/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.



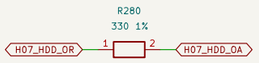






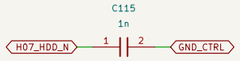


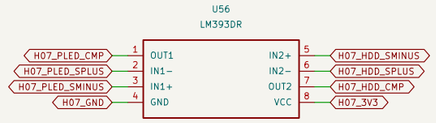










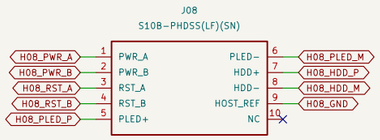


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 07: isolated controls / LEDs (3/3)/
File: host07_3.kicad_sch

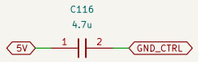
Title: Host 07: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 30/39

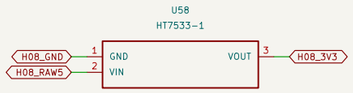
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.









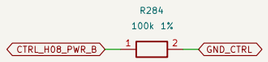








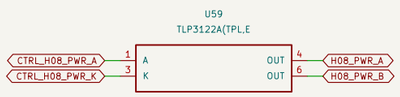






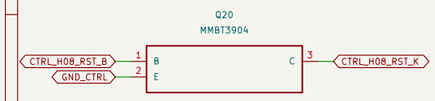






















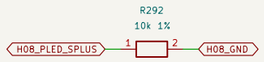


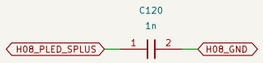


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 08: isolated controls / LEDs (1/3)/
File: host08_1.kicad_sch
Title: Host 08: isolated controls / LEDs (1/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 31/39

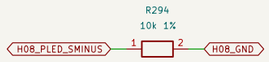
All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

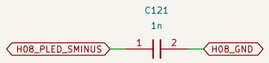




























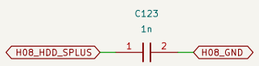




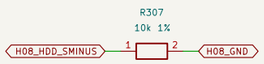


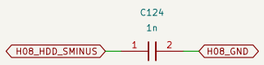


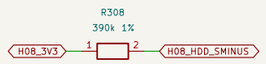










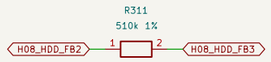


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 08: isolated controls / LEDs (2/3)/
File: host08_2.kicad_sch
Title: Host 08: isolated controls / LEDs (2/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 32/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.









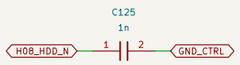












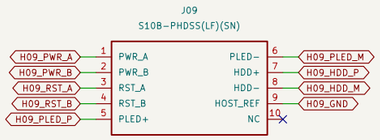


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 08: isolated controls / LEDs (3/3)/
File: host08_3.kicad_sch

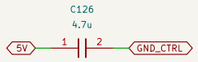
Title: Host 08: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 33/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.







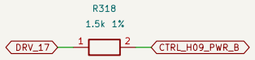












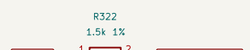




























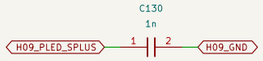


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 09: isolated controls / LEDs (1/3)/
File: host09_1.kicad_sch
Title: Host 09: isolated controls / LEDs (1/3)
Size: A3 | Date: | **Rev: B.1**
KiCad E.D.A. 10.0.0 | Id: 34/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

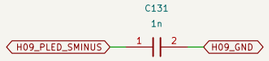


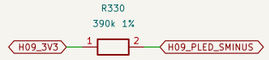


















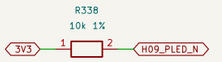








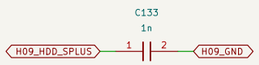




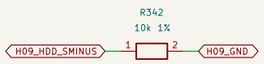


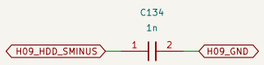














SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 09: isolated controls / LEDs (2/3)/
File: host09_2.kicad_sch

Title: Host 09: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 35/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.











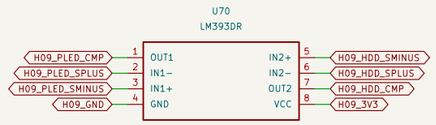










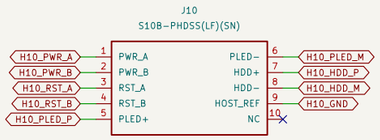


SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 09: isolated controls / LEDs (3/3)/
File: host09_3.kicad_sch

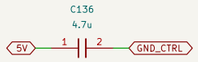
Title: Host 09: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 36/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.







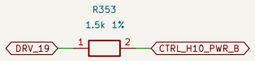










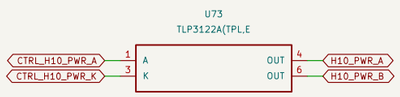






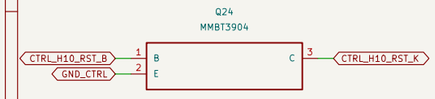


























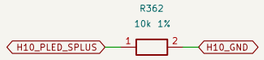
SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 10: isolated controls / LEDs (1/3)/
File: host10_1.kicad_sch

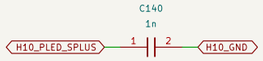
Title: Host 10: isolated controls / LEDs (1/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 37/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.

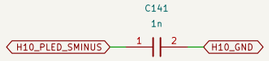


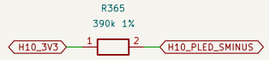
















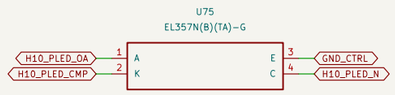










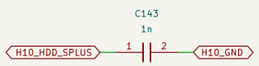




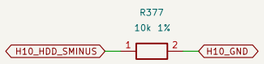


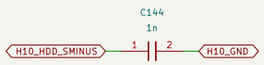














SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 10: isolated controls / LEDs (2/3)/
File: host10_2.kicad_sch

Title: Host 10: isolated controls / LEDs (2/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 38/39

All HOST_REF and ESD grounds on this sheet belong only to this host.
Case buttons and LEDs remain straight-through on passive adapter. No host power is supplied on the cable.
LED input envelope: 0..5 V common-mode; OFF differential <=0.2 V; ON >=1.4 V. Qualify actual motherboard.
LM393 inputs tolerate positive voltage with VCC=0; 100k series limits signal loading. Not a universal F_PANEL map.





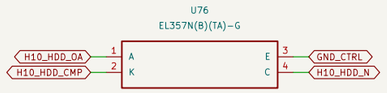






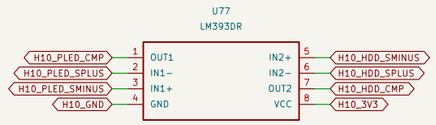












SCHEMATIC REVIEW — not a fabrication release
Rack ATX Controller / ddasdkimo
Sheet: /Host 10: isolated controls / LEDs (3/3)/
File: host10_3.kicad_sch

Title: Host 10: isolated controls / LEDs (3/3)

Size: A3	Date:	**Rev: B.1**
KiCad E.D.A. 10.0.0		Id: 39/39